

Brendan C. Wallace

Atmospheric Data Scientist

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Education

University at Albany, State University of New York

Ph.D. Atmospheric Science, M.S. Atmospheric Science

Aug 2016 - Dec 2022

Research Areas: Numerical modeling, climate dynamics, land surface-atmosphere interactions

Western Illinois University

B.Sc. Meteorology

Aug 2012 - May 2016

Research Areas: Geospatial statistics, GIS, high-performance computing

Work Experience

Atmospheric Data Scientist

Mar 2024 - Present

Leidos - Earth Intelligence

- Contributed to and led technical development of several DoD research proposals geared towards addressing needs in environmental monitoring and prediction
- Led physics-informed AI research initiatives evaluating the capability of deep learning frameworks at predicting snow distribution across mountain basins
- Processed, reformatted, and optimized multiple large-scale geospatial datasets, primarily remotely sensed multispectral data and NWP model output, for ML workflows
- Authored several technical documents including peer-reviewed papers, activity reports, and capability documents disseminated to a broad audience of varying backgrounds and skill levels

Joint Appointee Postdoctoral Fellow

Sep 2022 - Mar 2024

Northern Illinois University and Argonne Natl. Lab

- Analyzed over 1 PB of output from multi-year convection permitting regional climate simulations and reformatted key variables into analysis-ready format for efficient processing at scale in an HPC environment
- Applied novel decomposition and classification frameworks to better understand projected changes in precipitation from atmospheric model output
- Updated pre-existing Python analysis workflows to better integrate Xarray and Dask
- Published two peer-reviewed articles and presented work at conferences

Graduate Research Assistant

Aug 2016 - Sep 2022

Research Foundation - University at Albany

- Performed and analyzed decadal convection-permitting regional climate simulations with WRF that was recognized and awarded by both the university and the department
- Applied linear regression analysis for calibration of air pollution monitoring sites for NYSEDA
- Published three peer-reviewed articles and presented at multiple conferences

Skills

Programming Languages

Python, Fortran, Bash, SQL

Modeling and Computing

HPC, Slurm, PBS, MPI, WRF, MPAS

Data Science

Deep learning (PyTorch), Uncertainty quantification, Feature importance

Software

LaTeX, ESRI Products, QGIS, Unix, Git, Docker

Publications

- **Wallace, B.**, Haberlie, A. M., Gensini, V. A., Ashley, W. S., & Michaelis, A. C. (2025). Cause and Characteristics of Changes in Mesoscale Convective Systems within a Convection-Permitting Regional Climate Model. *Journal of Climate*, 38(2), 461–479. <https://doi.org/10.1175/jcli-d-24-0251.1>
- Gensini, V. A., Ashley, W. S., Michaelis, A. C., Haberlie, A. M., Goodin, J., & **Wallace, B. C.** (2024). Hailstone size dichotomy in a warming climate. *Npj Climate and Atmospheric Science*, 7(1). <https://doi.org/10.1038/s41612-024-00728-9>
- Haberlie, A. M., **Wallace, B.**, Ashley, W.S., Gensini, V.A., Michaelis, A.C. (Submitted). Mesoscale Convective System Activity in the United States Under Intermediate and Extreme Climate Change Scenarios.
- **Wallace, B.**, & Minder, J. R. (2024). The sensitivity of the North American Monsoon to Gulf of California Sea surface temperatures. In *Climate Dynamics*. Springer Science and Business Media LLC. <https://doi.org/10.1007/s00382-023-07057-2>
- **Wallace, B.**, Haberlie, A. M., Ashley, W. S., Gensini, V. A., & Michaelis, A. C. (2023). Decomposing the Precipitation Response to Climate Change in Convection Allowing Simulations Over the Conterminous United States. In *Earth and Space Science* (Vol. 10, Issue 12). American Geophysical Union (AGU). <https://doi.org/10.1029/2023ea003094>
- **Wallace, B.**, & Minder, J. R. (2023). The North American Monsoon precipitation response to climate warming at convection-permitting scales. In *Climate Dynamics*. Springer Science and Business Media LLC. <https://doi.org/10.1007/s00382-023-06920-6>
- **Wallace, B.**, & Minder, J. R. (2021). The impact of snow loss and soil moisture on convective precipitation over the Rocky Mountains under climate warming. *Climate Dynamics* (Vol. 56, Issues 9–10, pp. 2915–2939). <https://doi.org/10.1007/s00382-020-05622-7>
- Deng, Y., **Wallace, B.**, Maassen, D., Werner, J. (2016). A Few GIS Clarifications on Tornado Density Mapping. *Journal of Applied Meteorology and Climatology*, 55(2), 283-296.

Awards

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| • 2023 Narayan R. Gokhale Distinguished Research Scholarship Award | 2023 |
| • University at Albany Distinguished Doctoral Dissertation Award | 2023 |
| • 1st Place Student Oral Presentation, 18 th Conference on Mtn Met. | 2018 |
| • Arthur G. Tillman Scholarship, Western Illinois Univ. Dept. of Geog. | 2015 |
| • Presidential Scholarship, Western Illinois Univ. | 2012-2016 |

Professional Affiliations

- American Meteorological Society